

Insurance Cost Analysis

```
insurance <- read.csv("data/medical_insurance.csv")  
names(insurance)
```

[1] "person_id"	"age"
[3] "sex"	"region"
[5] "urban_rural"	"income"
[7] "education"	"marital_status"
[9] "employment_status"	"household_size"
[11] "dependents"	"bmi"
[13] "smoker"	"alcohol_freq"
[15] "visits_last_year"	"hospitalizations_last_3yrs"
[17] "days_hospitalized_last_3yrs"	"medication_count"
[19] "systolic_bp"	"diastolic_bp"
[21] "ldl"	"hba1c"
[23] "plan_type"	"network_tier"
[25] "deductible"	"copay"
[27] "policy_term_years"	"policy_changes_last_2yrs"
[29] "provider_quality"	"risk_score"
[31] "annual_medical_cost"	"annual_premium"
[33] "monthly_premium"	"claims_count"
[35] "avg_claim_amount"	"total_claims_paid"
[37] "chronic_count"	"hypertension"
[39] "diabetes"	"asthma"
[41] "copd"	"cardiovascular_disease"
[43] "cancer_history"	"kidney_disease"
[45] "liver_disease"	"arthritis"
[47] "mental_health"	"proc_imaging_count"
[49] "proc_surgery_count"	"proc_physio_count"
[51] "proc_consult_count"	"proc_lab_count"
[53] "is_high_risk"	"had_major_procedure"

What proportion of insurance cost variance is attributable to modifiable lifestyle choices versus baseline demographic and socioeconomic risk factors?